MACHINE LEARNING | ALGORITHM ATLAS

12 methods • learning approach • application domain • practical use case

Linear Regression

Supervised | Tabular

Fits a linear relationship to predict continuous values.

USE CASE: Forecast daily transit boardings.

Decision Tree

Supervised | Tabular

Splits records using feature-based decision rules.

USE CASE: Classify transit stops by performance.

Random Forest

Supervised | Tabular

Aggregates predictions from many randomized trees.

USE CASE: Predict paratransit trip demand.

Gradient Boosting

Supervised | Tabular

Adds models sequentially to correct prior errors.

USE CASE: Estimate corridor travel times.

Support Vector Machine

Supervised | Tabular / NLP

Finds a margin-maximizing classification boundary.

USE CASE: Classify service requests.

K-means

Unsupervised | Tabular

Groups similar observations around cluster centers.

USE CASE: Segment transit rider patterns.

Learning paradigms are not identical to application domains; generative models may use other training objectives.

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PCA

Unsupervised | Tabular / Vision

Projects features onto directions of maximum variance.

USE CASE: Summarize correlated accessibility indicators.

CNN

Usually supervised | Computer vision

Uses learned filters to extract spatial image patterns.

USE CASE: Detect bus stops in street imagery.

BERT

Self-supervised + fine-tuned | NLP

Learns bidirectional text representations with attention.

USE CASE: Classify public comments by topic.

GPT / autoregressive transformer

Self-supervised + aligned | NLP / Generative AI

Predicts the next token to generate coherent text.

USE CASE: Draft summaries of planning documents.

GAN

Adversarial generative | Generative AI / Vision

Trains a generator against a discriminator.

USE CASE: Create synthetic training imagery.

Diffusion model

Generative denoising | Generative AI / Vision

Learns to reverse noise to synthesize samples.

USE CASE: Generate concept illustrations.

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